

Yahya Masri

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EDUCATION

George Mason University

May 2025 - May 2027

Bachelor of Science, Computer Science (GPA: 4.0)

- **Coursework:** Data Structures, Formal Methods and Models, Introduction to Low-level Programming

EXPERIENCE

Lillup | *AI Software Engineer*

Jan 2026 - Present

- Built a **local RAG pipeline** that generates **source-cited** educational content using ChromaDB, LangChain, and GBNF-constrained generation via llama-cpp-python on quantized CPU models, resulting in consistently source-grounded output and reduced manual content curation effort.
- Benchmarked three **on-device SLMs** across 20 queries each on parse reliability and hallucination rate, selecting Qwen3 0.6B at an average latency of **4.21s** with zero parse failures.
- Implemented **retrieval-stage grounding enforcement** through **cosine distance threshold filtering**, achieving **100% precision** and **96% accuracy** across 100 test queries, eliminating hallucinated output.

George Mason University | *Teaching Assistant*

Jan 2026 - Present

- Supported a **292-student** CS 262 course by answering **conceptual and programming questions** on Piazza, which improved student satisfaction and helped maintain high assignment completion rates
- Explained **low-level programming concepts** clearly and concisely to help students understand course material and assignment requirements.

NSF Spatiotemporal Innovation Center, George Mason University | *Undergraduate Research Assistant*

Feb 2024 - Present

- Benchmarked **SLMs and SRLMs** on real-world Linux system logs, achieving **96% accuracy** with retrieval-augmented prompting to evaluate deployability for real-time digital twin monitoring systems.
- Achieved **91% accuracy** on conflict-text classification by comparing **BERT and GPT** using hyperparameter tuning and prompting methods such as **RAG, RAG-ICL, and zero-shot** approaches.
- Optimized **context-based location-extraction** performance to **90%** through evaluation of open-source LLMs using various prompting methods and aligned evaluation methods.
- Reduced manual GIS processing time by **90%** through **automated conflict-mapping** workflows using a **multi-agent system (MAS)** integrated with QGIS, Geocoding API, and Streamlit.

PROJECTS

Glimpse | <https://tryglimpse.dev/>

- Co-developed an **AI-powered product video startup** that transforms a one-sentence product description into a cinematic 12-second video using a **MAS** with **GPT-5.1** and **Sora-2-Pro**
- Led product research and positioning, **analyzing trends** across recent YC batches to inform Glimpse's **launch strategy** focused on storytelling, differentiation, and waitlist validation.
- Implemented the **core AI pipeline and backend**, orchestrating the GPT-5.1 agents to Sora-2-Pro through a **Python/OpenAI SDK** stack to generate structured prompts and production-ready video outputs.
- Collaborated across a 4-person team to ship a functional **end-to-end** product from concept to demo, integrating AI, video generation, and product UX into a unified experience, using **Git** version control.

SKILLS

- **Programming Languages:** Python, C, Java, LaTeX
- **Frameworks/Libraries:** LangChain, ChromaDB, FAISS, llama-cpp-python, Streamlit, pandas, NumPy, PyTorch, scikit-learn
- **Tools & Platforms:** Git, Linux, PostgreSQL, MongoDB, LM Studio, Ollama, Huggingface

SELECTED PUBLICATIONS

- **Masri, Y.**, Ma, E., Wang, Z., Rogers, J. and Yang, C..Benchmarking Small Language Models and Small Reasoning Language Models on System Log Severity Classification.*arXiv preprint arXiv:2601.07790*. <https://doi.org/10.48550/arXiv.2601.07790>
- **Masri, Y.**, Wang, Z., Srirenganathan Malarvizhi, A., Ahmed, S., Stover, T., Wong, D.W., Jiang, Y., Li, Y., Liu, Q., Bere, M. and Rothbart, D..Comparative Analysis of BERT and GPT for Classifying Crisis News with Sudan Conflict as an Example.*Algorithms*, 18(7), p.420. <https://doi.org/10.3390/a18070420>
- Wang, Z., **Masri, Y.**, Malarvizhi, A.S., Stover, T., Ahmed, S., Wong, D., Jiang, Y., Li, Y., Bere, M., Rothbart, D. and Pfoser, D..Optimizing context-based location extraction by tuning open-source LLMs with RAG.*International Journal of Digital Earth*, 18(1), p.2521786. <https://doi.org/10.1080/17538947.2025.2521786>